Faculty of Computer Science | Master of Data Science | Applied Quantitative Logistics

Genetic Speciation Algorithm for Addressing Combinatorial Optimization Problems:

Application to the Vehicle Routing Problem

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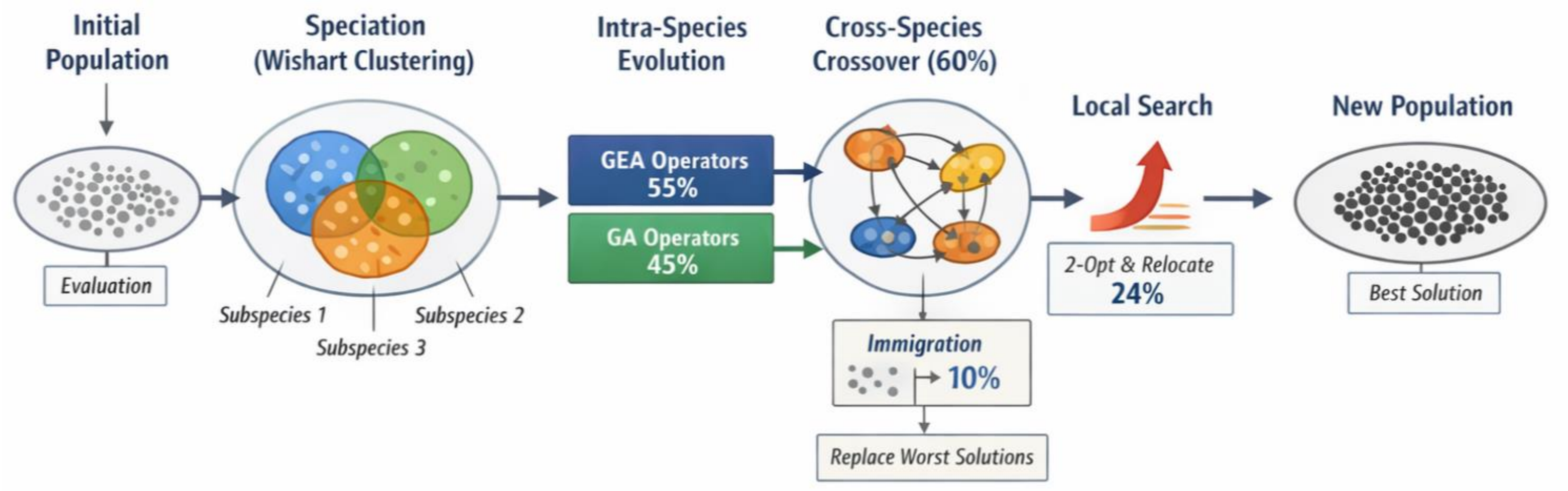
Genetic Speciation Algorithm



GEA Operators: 55%

- Scenario 1: 15%
- Scenario 2: 15%
- Scenario 3: 25%

GA Operators: 45%

Cross-Species Interaction: 60%

Immigration Rate: 10%

Speciation

- Method: Wishart
- Number of Species: 3

Local Search

- Applied to Top 40%
- Activation Rate: 60%
- Effective Ratio: ≈ 24%

Wishart-Based Speciation in GSA

Problem in Single Population

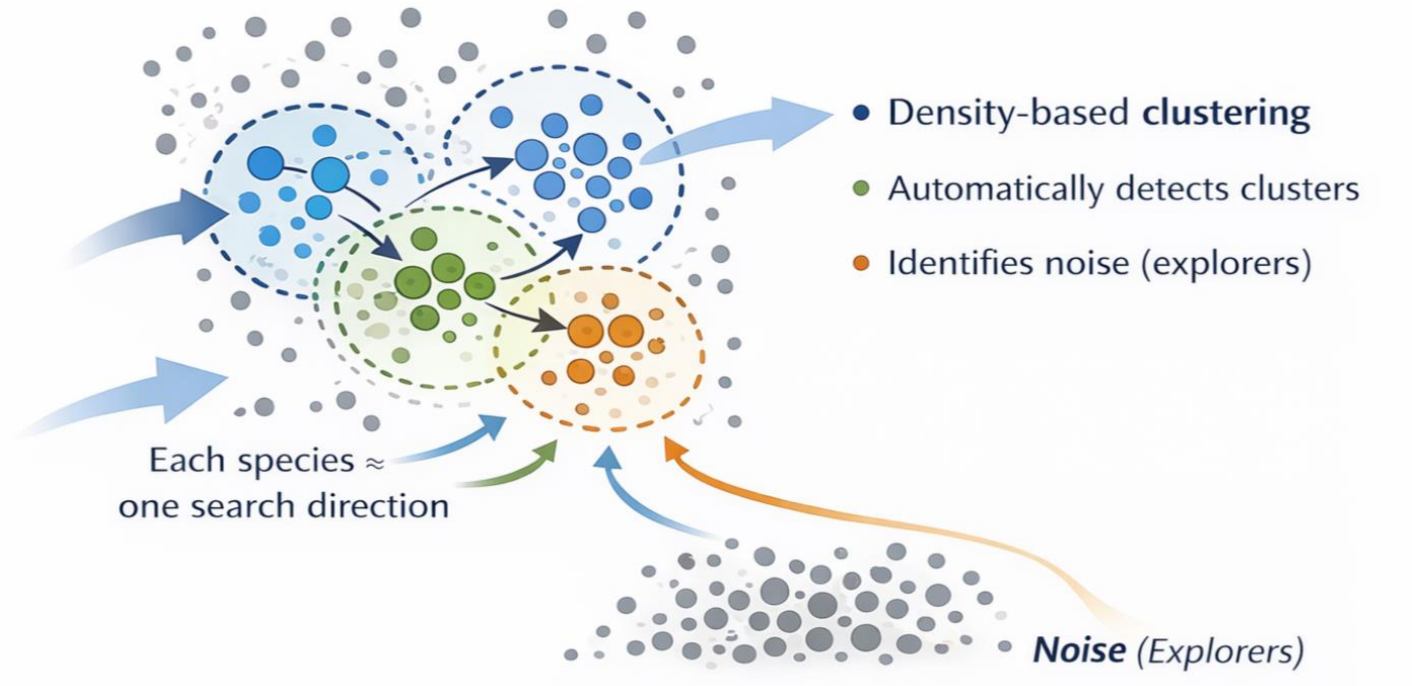
- Loss of diversity
- Premature convergence
- Dominance of a single structure

Wishart Clustering

- Density-based clustering
- Automatically detects clusters
- Identifies noise (explorers)

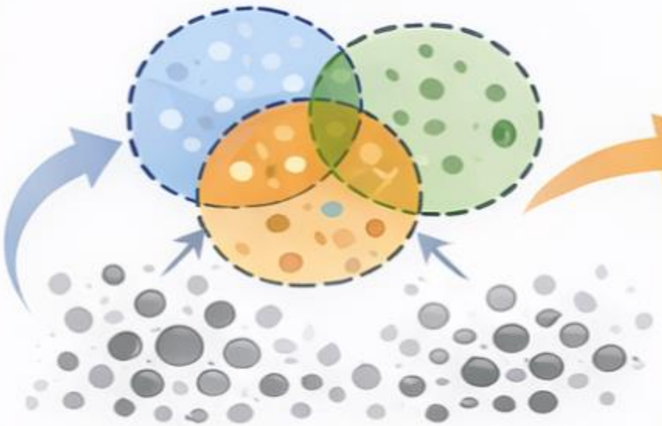
Why Wishart?

- No need to predefine cluster shapes
- Captures structural similarity
- Robust to noise



Enhancement Mechanisms in GSA

Cross-Species Interaction (60%)



Promotes information flow across species

- Exchange information across species
- Break isolation between subpopulations

Immigration (10%)

- Replace **worst individuals**
- Introduce random solutions
- Prevent premature convergence

Injects diversity into the population

- Replace **worst individuals**
- Introduce random solutions
- Prevent premature convergence

Local Search (≈24%)

- Applied to **top 40%** individuals
- Activation rate: 60%

Operators: 2-opt & relocate

Enhances solution quality via refinement

Performance Evaluation of GSA

Instances

- A-n32-k5
- A-n60-k9
- A-n80-k10

Algorithms

- GA
- GSA

Settings

- 10 runs per instance
- 300 generations
- Population size: 100